Milestone #2
CompSci 325 - Fall 2024

Ctrl Park 🚗

Team Members

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Tasks

The three areas we have selected to cover with tasks are:

1. Real-time parking availability
2. Parking spot reservation
3. Violation alerts.

These are key features to the application that highlight unique aspects that will be crucial for users to understand.

1. Real-Time Parking Availability

This feature allows users to view up-to-date information about which lots currently have open spaces. Map View helps users locate lots relative to their current position, whereas List View provides a scrollable format for browsing. Both views support search and filter functions, allowing users to find specific lot numbers or narrow down their results. Filter options include availability, location, permit type, walking distance, price, and accessibility.

Using Map View to find open parking spot:

1. Open application → sent to the home screen on ctrl park.
2. The home screen is the map view. Each lot is bordered and titled. Users can pinch and zoom to move around the map to view locations of parking lots.
3. Each parking lot has a color, which communicates availability status:
    - Green: Less than 40% of spaces are occupied.
    - Yellow: 40-70% of spaces are occupied.
    - Red: More than 70% of spaces are occupied.
4. Click on a lot to view spaces in the parking lot. Green indicates open spots and red indicates occupied.
5. Tap an open space to view details (spot ID, restrictions, time limits, last updated).

6. Tap “Navigate” for turn-by-turn directions to the lot entrance. Park and tap “I Parked” so the system marks the spot as occupied and starts your parking session/timer. (If a spot becomes unavailable en route) tap “Find Alternatives” to get the nearest open spots and reroute.

Searching by lot number on Map View:

1. Open the application → sent to the home screen on ctrl park.
2. Click on the search bar on the bottom controls.
3. Type in parking lot number.
4. Map automatically updates with every number typed. For example, if a user typed “4”, lots that start with “4” remained on the map. If a user typed “42”, only lot 42 remained on the map.
5. Delete search query for all the lots to appear.

Applying a filter to the Map View:

1. Open application → sent to the home screen on ctrl park.
2. Click on the “Filter” icon on the bottom controls → brings up the filter screen.
3. The filter screen is divided into filter criterias.
4. Click on the checkboxes or adjust the sliders that match the desired outcome.
5. Click “Cancel” if the user decided not to apply any changes.
6. Click “Apply” to apply all the selected filters → the filter screen disappears.
7. Map View updates to match the filter options. Parking lots that do not match do not appear on the map.
8. The number on the “Filter” icon indicates the number of filters applied.

Using List View to find open parking spot:

1. Open application → sent to the home screen on ctrl park.
2. Click on the “list view” button on the bottom controls → list view page slides up.
3. If the user wants to exit the list view, swipe the page down or click on the “X” button on the top right of the list view page.
4. List view page shows all the parking lots ordered numerically. Each lot is contained in a card view. Additionally, each card contains information about address, open spaces, total spaces, and allowed permit type. The card's border color communicates availability status:
    - Green: Less than 40% of spaces are occupied.
    - Yellow: 40-70% of spaces are occupied.
    - Red: More than 70% of spaces are occupied.
5. Users can click on their preferred parking lot → list view page slides down and map view narrows onto that parking lot.
6. Tap an open space to view details (spot ID, restrictions, time limits, last updated).
7. Tap “Navigate” for turn-by-turn directions to the lot entrance.

Park and tap “I Parked” so the system marks the spot as occupied and starts your parking session/timer.
(If a spot becomes unavailable en route) tap “Find Alternatives” to get the nearest open spots and reroute.

Searching for lot number on List View:

1. Open application → sent to the home screen on ctrl park.
2. Click on the “list view” button on the bottom controls → list view page slides up.
3. Click on the search bar at the top of the list view page.
4. Type in parking lot number.
5. Map automatically updates with every number typed. For example, if a user typed “4”, lots that start with “4” remained on the map. If a user typed “42”, only lot 42 remained on the map.
6. Delete search query for all the lots to appear.

Applying a filter to the List View:

1. Open application → sent to the home screen on ctrl park.
2. Click on the “list view” button on the bottom controls → list view page slides up.
3. Click on the “Filter” icon on the top right → brings up the filter screen.
4. The filter screen is divided into filter criterias.
5. Click on the checkboxes or adjust the sliders that match the desired outcome.
6. Click “Cancel” if the user decided not to apply any changes.
7. Click “Apply” to apply all the selected filters → the filter screen disappears
8. List View updates to match the filter options. Parking lots that do not match do not appear on the list.
9. The number on the “Filter” icon indicates the number of filters applied.

Outcome: After using this task, users gain a clearer understanding of current parking availability and can narrow down to the lot that best fits their need. They can also locate a specific parking spot within a lot and navigate directly to it.

2. Parking Spot Reservation

This feature allows users to reserve an available parking space before arriving on campus for a set period. Thus promoting fairness and predictability by ensuring that students who plan ahead can secure convenient parking without competing in real-time. Additionally, they will be able to modify or cancel reservations, so that in the event reservations are no longer needed, spots are available for other users without penalty.

Using Reservation Features:

1. Open application → sent to home screen on ctrl park

2. Click on reservations button on bottom right of menu bar → brings to main reservations screen

**** From here there are three paths the user can take to interact with reservations****

Creating New Reservation:

1. Click “Create New Reservation” button → brings to new reservation screen
2. Click on multifunctional calendar → navigate between different months and dates
3. Click on months and days → selects date of reservation
4. Drag clock arms → choose time of reservation
5. Click between AM + PM → change to AM or PM reservation
6. Select “Cancel” → backs out user from creating reservation
7. Select “Ok” in bottom right of clock → brings to reservation preferences screen
8. Adjust days, hours, minutes using arrow buttons → increments length of reservation
9. Click between permit types (“Commuter”, “Resident”, “Guest”) → selects type of permit
10. Check off options in checkbox → selects 0-4 preferences users can choose from
11. Click “Search Availability” button → searches available lots
12. Click “Back” button → brings user back to preferences screen
13. Click on any of highlighted lots → shows lot details
14. Click “Back” button → brings user back to screen showing highlighted lots
15. Click “Reserve” button → moves user to payment and summary screen
16. Click “Modify” button → brings user back to lot details screen
17. Choose plate numbers in drop down menu → selects the plate # to reserve for
18. Click between payment methods (“Apple Pay”, “Credit Card”, “Venmo”) → alternates between payment methods available
19. Click “Use a different Credit Card” → Allows user to choose a different payment card
20. Click “Confirm” button → brings user to success screen and books the user’s parking reservation

Checking into Reservation:

1. Click “Check-In to Existing Reservation” → brings user to upcoming reservation screen
2. Click “Navigate to Lot” → redirects user to native navigation app with location information for parking lot
3. Click “Check-In / I Arrived” → brings user to a welcome screen for lot and runs a countdown for reservation time

Modifying / Cancelling Existing Reservations:

1. Click “Cancel / Modify Reservation” → brings user to screen with current reservations

2. Click on button for reservation user wants to modify → brings user to reservation details
3. Click “Modify” or “Cancel” reservation → brings user to a screen for either modifying or cancelling reservation
    a. Click “Modify” → brings user to modification screen
        i. Choose between “Modify Start Time” or “Modify End Time” → will modify whether user is altering start / end time
        ii. Adjust days, hours, minutes using arrow buttons → increments length to reduce or add to reservation
        iii. Click “Confirm” → brings user to modification summary screen
        iv. Choose plate numbers in drop down menu → user can change plate the reservation is for
        v. Click “Modify” → backs user out to modification screen
        vi. Click “Confirm” → brings user to success screen confirming modification
    b. Click “Cancel” → brings user to an “are you sure” screen for cancelling
        i. Click “Go Back” → brings user back to modify / cancel
        ii. Click “Yes, Cancel” → brings user to a cancelled success screen

Outcome: From the reservation task, the user will have been able to successfully create, modify, cancel, or check-in to their desired parking lot reservation.

3. Violation Alerts and Policy Notifications

This feature provides notifications when users receive a parking ticket, if time limits are approaching on parking spots, as well as updates on any temporary rule lot closures for things such as tailgates. For this task there will be 3 different types of notifications the user will receive and paths to follow.

Turning on Notifications:

1. Open application → sent to the home screen of ctrl park
2. From the home screen select setting icon in top right corner
3. Once in settings select “Notifications”
4. On the notifications page turn on all notifications and select SMS, email or both as a channel to receive notifications
5. Close the application and go to home screen

∗From here there are three different paths to follow regarding different notifications∗

Lot Closure:

1. From the home screen, the user receives a notification “LOT CLOSURE: Lot 12...” click on the button at the bottom of the notification that says “Click here for more information”

2. On the home screen of the application there will be an exclamation point icon on Lot 12 and it is red, click on the icon
3. Read the announcement and then close the alert

Adding Time to Reservation:

1. From the home screen, the user receives a notification saying, “You have 15 minutes remaining...” at the bottom of this alert, click the button that says, “click here to add more time”
2. User is directed to the home screen where there is a pop-up with a clock with a banner saying “Add more time”
3. Choose the amount of time to add to the parking reservation and click “ok” in the bottom right corner of the pop-up
4. A confirmation option will pop-up insuring that all the information is correct, telling you how much it will cost → click “yes” if all the information is correct
5. Once confirmed there will be a screen saying, “you have successfully added... to your reservation” → close this popup

Parking Ticket Notification:

1. From the home screen, the user receives a notification saying “your vehicle with plates... has received a ticket in lot...” at the bottom of the notification, click the button that says “click here for more information” → it will take you to the personal account page
2. On the personal account page, click parking tickets
3. Select the ticket that you would like to pay, there will be a red icon that says “pay citation”, click it
4. A pop-up will come up with a picture of the car when it got ticketed and more information → click pay ticket

Outcome: After the user completes this set of tasks they will have a general understanding of the layout of the application, the settings tab, the personal user tab, and how to pay parking tickets. Notifications will also be on so in the future the user will be updated on all information.

Low-Fidelity Prototypes

1. Real-Time Parking Availability

The Real-Time Parking Availability prototype was developed using Balsamiq. We utilized built-in controls such as buttons, scrollbars, search bar, icons, etc. This task allows the user to search for and navigate to a desired parking spot with ease. Users can choose between map view and list view, each with its own strengths and weaknesses depending on user preferences and needs. During the design process, we realized that users may

want a way to quickly locate parking lots that meet specific criterias. Hence, we had included filtering and searching functionalities in both views.

Strengths:

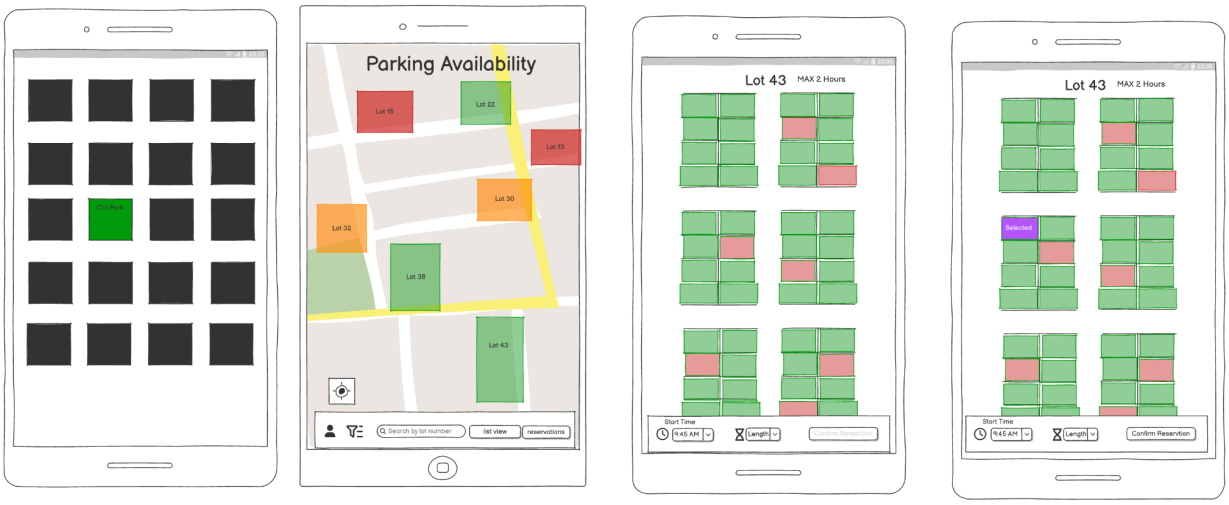
- We used colors to visually communicate the parking availability, enabling users to quickly scan and find suitable parking lots.
- The prototype offers two viewing options. Map view allows users to explore parking lots relative to their position, and List view, which provides a scrollable format where users can scan and compare.
- Consistent filtering and searching functionalities across the two views.

Weaknesses:

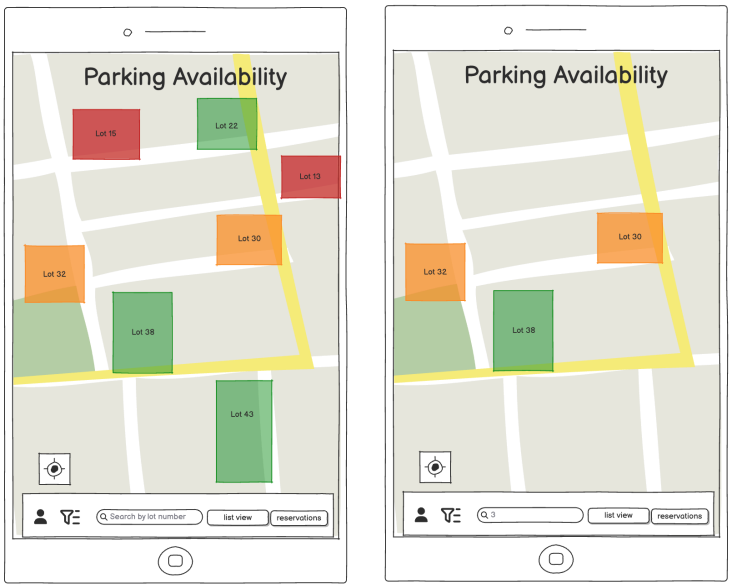
- Potential confusion between the two views. Users might interpret Map View and List View as entirely different functionalities rather than alternate ways to access the same information.
- Limited information about a parking lot.

Enter Map View and find a parking spot:

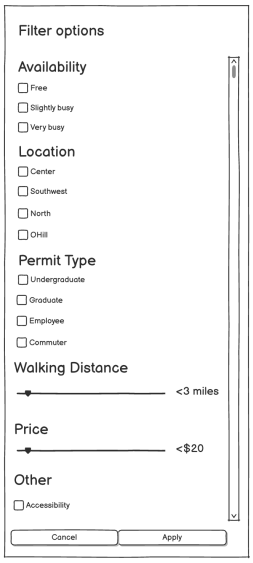
Open Application → Map View appears → Click on a parking lot → Click on a space



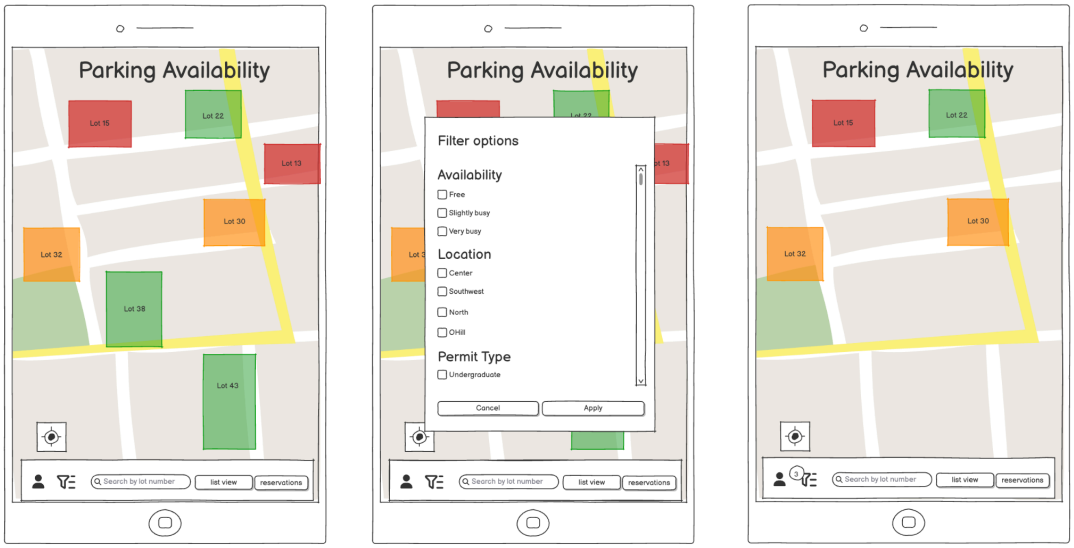
Search for a lot number on Map View:
Click on search bar → Type in lot number → Map view updates to match search query



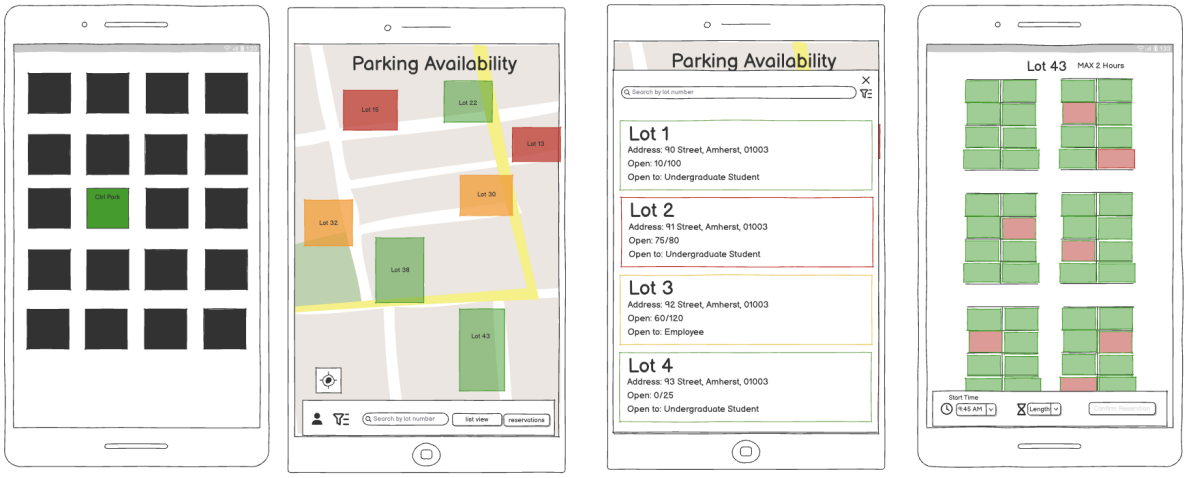
Filter page:
The page that appears whenever user click on a filter icon



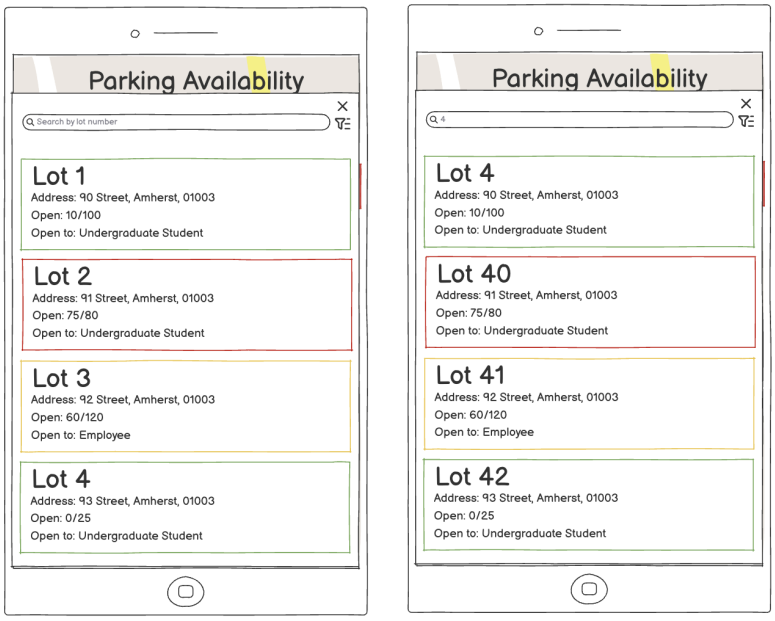
Apply a filter on Map View:
Click on the “Filter” icon → Apply the desired outcome → Click “Apply” → Map view updates to match filters



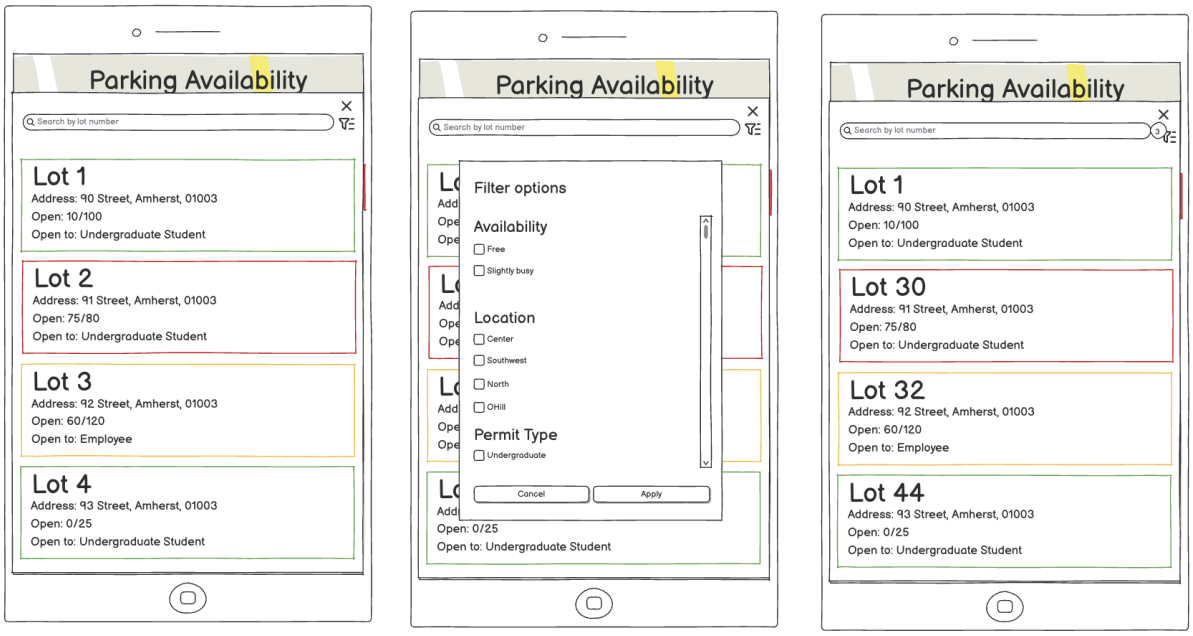
Enter List View and find a parking spot:
Open Application → Click on the “list view” button → List view page slides up → Scroll through the list to find parking lot → Click on a parking lot → Click on a space



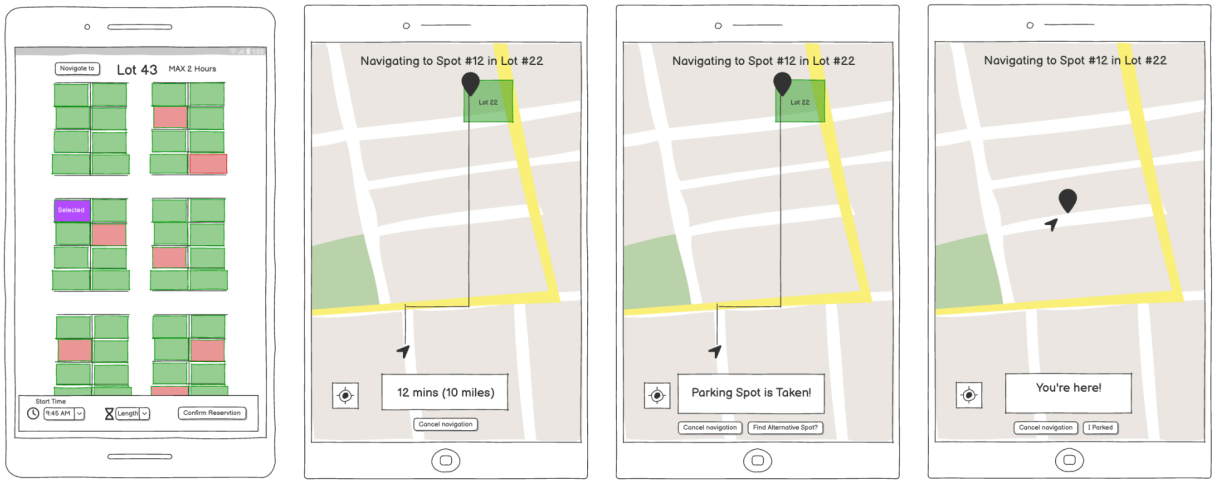
Search for a lot number on List View:
Click on search bar → Type in lot number → List view updates to match search query



Apply a filter on List View:
Click on the "Filter" icon → Apply the desired outcome → Click "Apply" → List view updates to match filters



Navigate to parking spot:
Click on preferred parking spot → Click on “Navigate to” → Shows direction to that parking spot → Click “I Parked” once in that parking spot



2. Parking Spot Reservation

The reservation low-fidelity prototype was made using Balsamiq. It was entirely created using the buttons, icons, boxes, dropdowns, etc. available on the software. The reservation workflow allows users to schedule, check-in, modify, and cancel reservations. These interactions are supported through multiple screens and buttons for each of the available tasks. During the creation of the low-fidelity prototype, we realized that all of these tasks were more in-depth than the surface value of simply saying a task should get done. Once it got to creating each of the screens there were multiple factors that came into play to allow the user functionality to get the task of reservations done. This workflow is presented below:

Strengths:

- Wide user control. In nearly every screen there is the ability for the user to back out of the actions and return to where they were.
- Confirmation screens. Provide user feedback so they know that their action has had an effect and completed.
- Consistent visual and interaction design. The controls throughout the reservation process replicate the same actions and buttons for easy learnability.

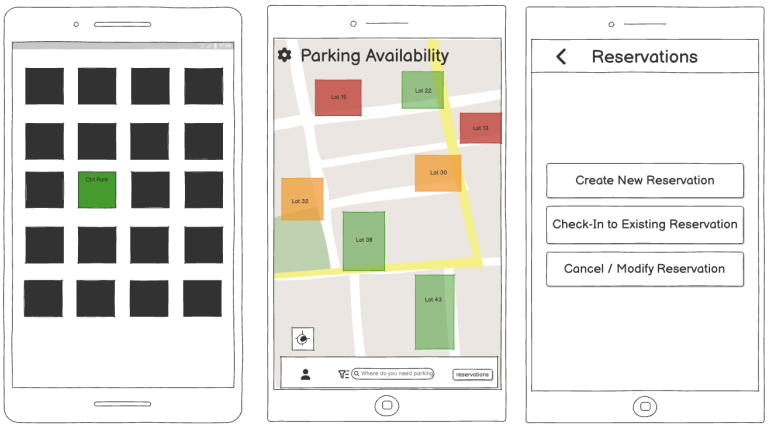
Weaknesses:

- Excessive click and step count. While creating the prototype we found that the process for creating the reservation may have been drawn out and could be simplified.
- Too much info on certain screens. Some screens such as the summary screen and the preferences screen may have too many options.

- Lack of shortcuts. Although the step-by-step process is an intuitive workflow, it might be beneficial to have other ways to navigate through the reservation windows.

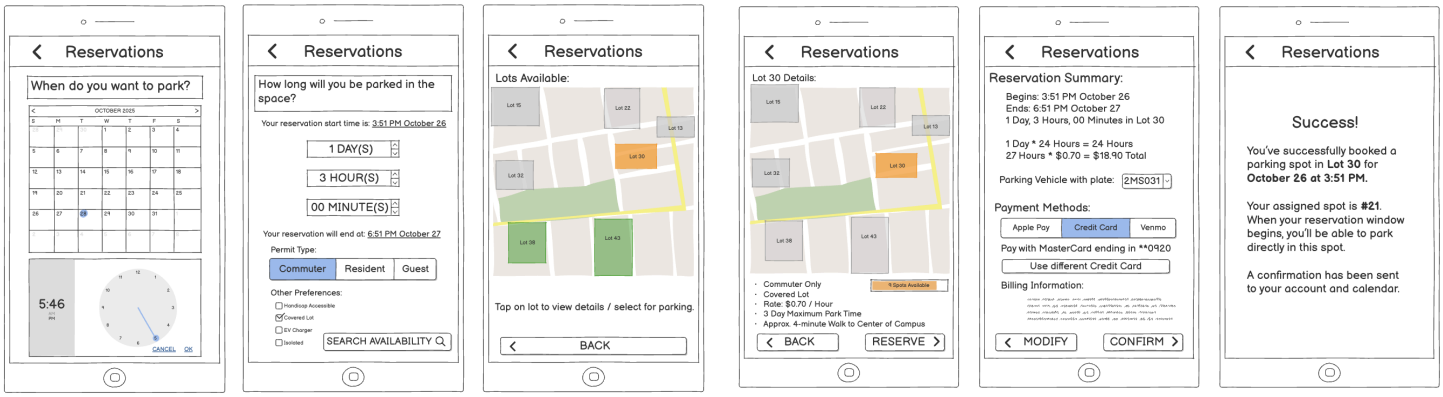
Enter Reservation screen:

Open Application → Open Reservations → Select Reservation Options



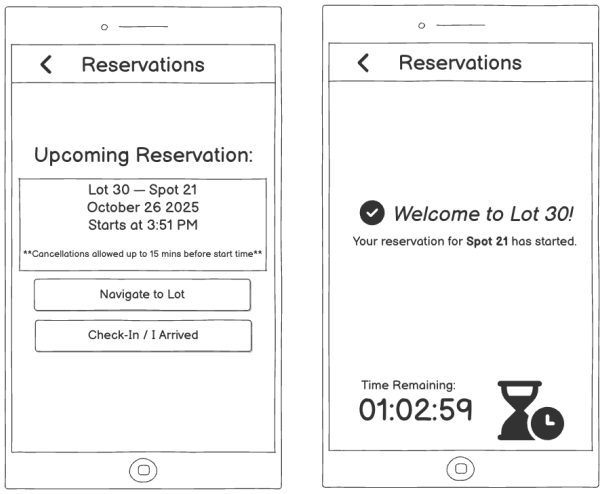
Create New Reservation:

Choose Reservation Day → Choose Time and Preferences → View Available Lots → View Lot Details → Reservation Summary → Success Screen



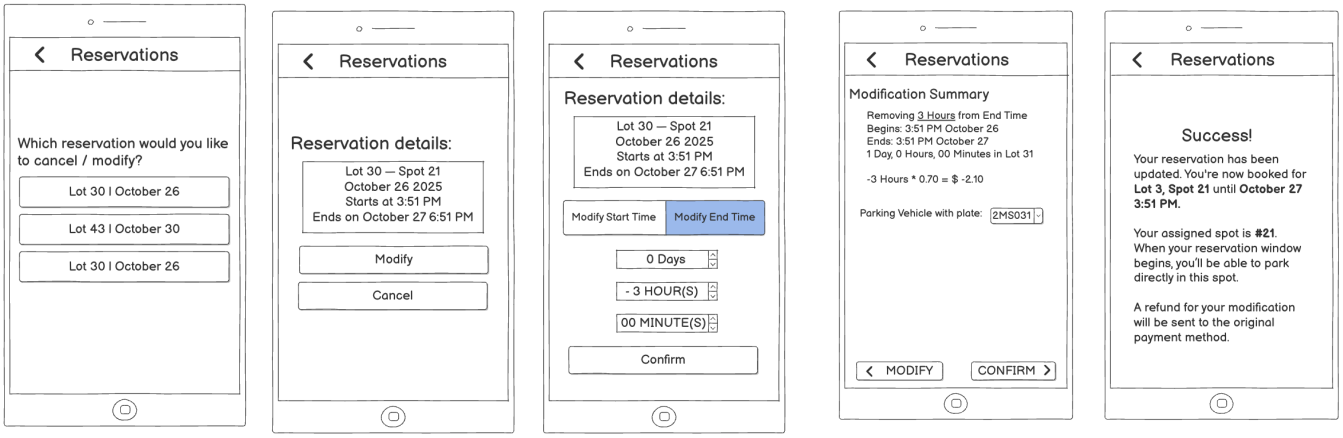
Check-In to Existing Reservation:

Reservation Information → Reservation Checked In with Countdown



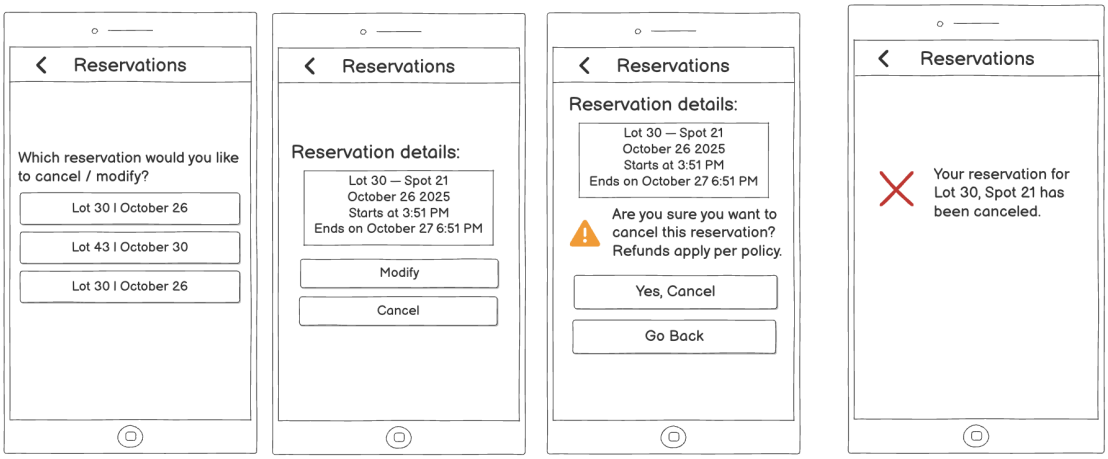
Modify Reservation:

Choose Reservation to Change → Modify or Cancel → Modify Reservation Length → Modification Summary Screen → Successfully Modified Screen



Cancel Reservation:

Choose Reservation to Change → Modify or Cancel → Choose to Cancel → Cancel Confirmation Screen



3. Violation Alerts and Policy Notifications

This prototype was created using Balsamiq to illustrate the interaction flow and key screens for notifications, lot closures, adding time, and paying tickets. The prototype for this task helps users turn on notifications then guides them through three actions based on possible alerts they may receive. This process helped us understand that it is crucial to be clear and use universal symbols as well as using layouts and organization that will be familiar to users. It also helped narrow down what notifications users will need to receive and helped focus the thought process.

Strengths:

- We used visual consistency, symbols are generally universal and common screens make it easy for users to interact with the system using intuition.
- The alerts clearly direct users to where they need to go and what they need to do leaving little room for error.
- There is visual feedback for when actions are completed, like the success screen for adding more time to a reservation.

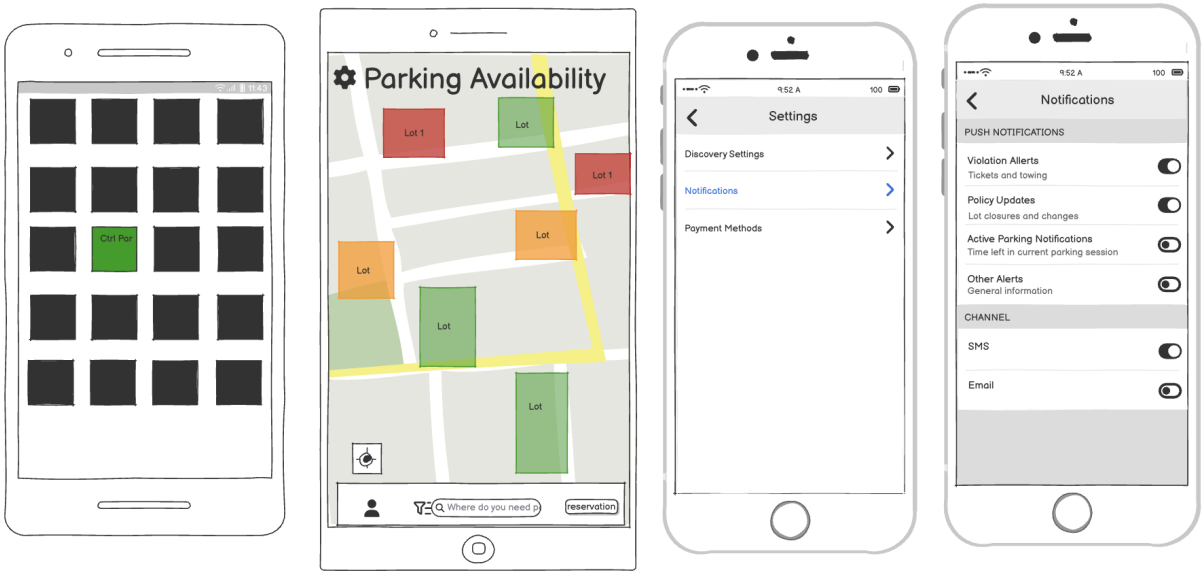
Weaknesses:

- There is a lack of context on how notifications are received, users turn on notifications and select the channel by which to receive them, then the prototype does not reflect this action, it just shows alerts.

- There could be more steps for the lot closure sequence that assists users in finding a place to relocate their car during the closures.
- For paying parking tickets, there could be a few more steps to actually go through with the process of paying the ticket.
- We have no room for error or error messages if something goes wrong.

Turning on Notifications:
This prototype will assist users in turning on notifications and selecting the way in which they would like to receive them.

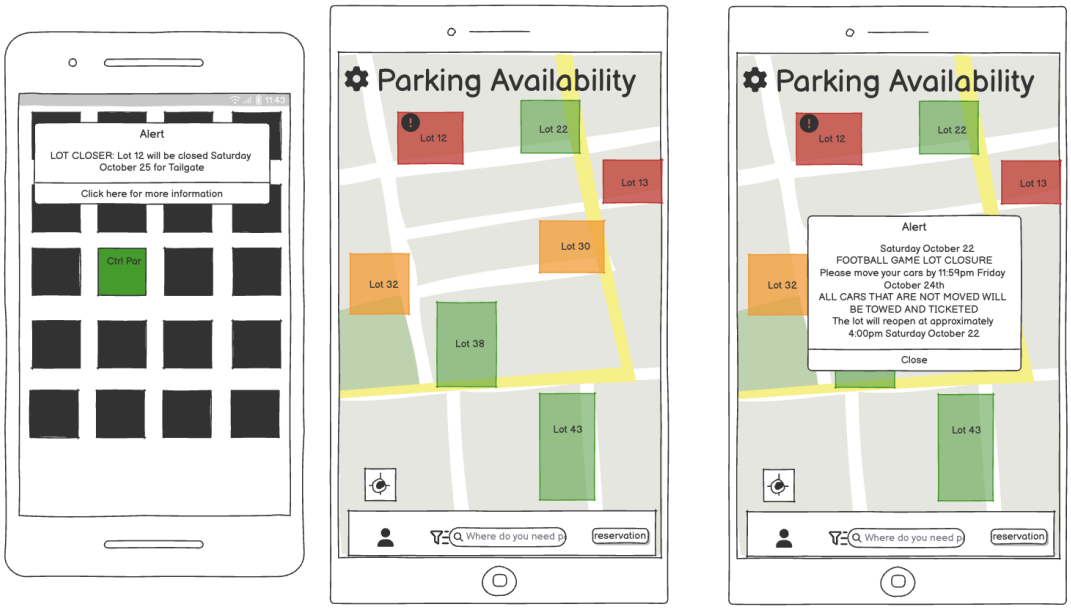
Open Application → Open Settings → Select Notifications → Turn on Notifications



Lot Closure:

This prototype will notify users of a parking lot closure and direct them to the app where they can receive more information about the closure.

Receive and Click on Alert → Select Lot With Warning Icon → View Alert for Lot



Add Time to Reservation:

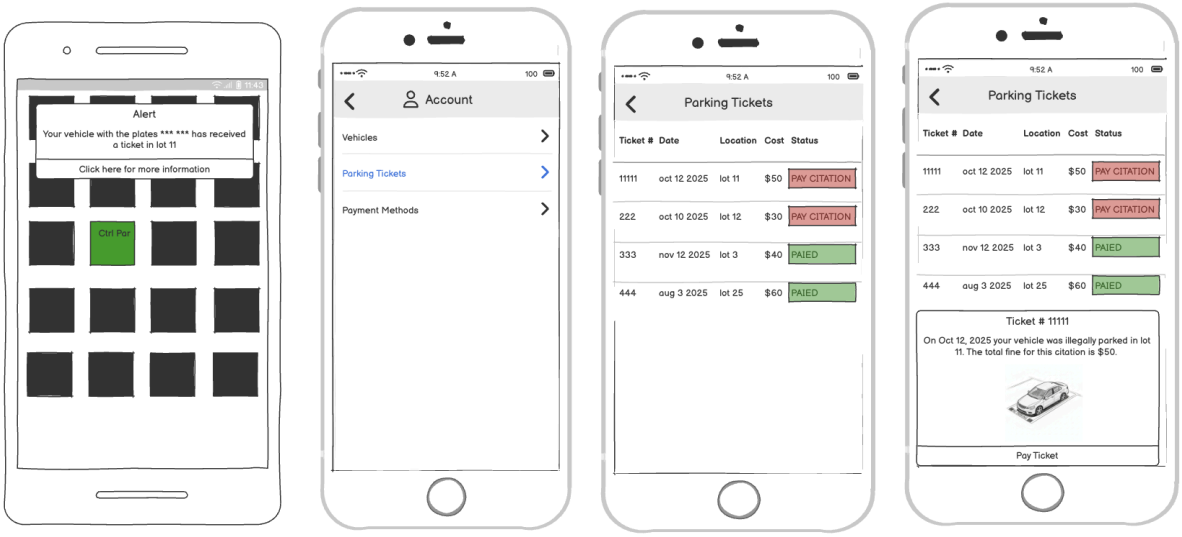
This prototype will notify users that they are low on time for their reservation and assist them in adding more time.

Click on Alert → Select Amount of Time to Add → Verify Information → Success!

Parking Ticket Notification:

This prototype will walk users through the process of receiving a parking ticket and paying said ticket.

Click on Alert → Select Parking Tickets → Select the Ticket to Pay → Pay Ticket



Participation

Louise

Completed all sections regarding the *Violation Alerts and Policy Notifications* task. Created the steps to follow in the task and fully designed the low-fidelity prototype. Formatted report and assisted others with peer review. Encouraged group members to complete tasks before our balsamiq trial runs out!

Nam

Created the workflow and prototypes for List View, lot navigation, filtering, and searching features in *Real-Time Parking Availability* task. Specified the strengths and weaknesses of this task.

Erik

Completed the map interface.

Kaylin

Created the Balsamiq wireframe for the group to collaborate on. Created the entire low-fidelity prototype for the *Parking Spot Reservation* on Balsamiq. Defined all of the tasks involved for reservation creation, modification, cancellation, and checking-in. Identified strengths and weaknesses for the reservation workflow and described its creation process.